

Decent Coaching Classes

IX CBSE

Mind Your Math - V

ANSWER KEY

Marks: 30

Time: 1 hr.

Section A – Multiple Choice Questions (1 mark each)

[6]

1. Answer: (B) 10 units. [Distance = $\sqrt{((-6)^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$, by the Pythagorean theorem]
2. Answer: (C) -2 and 5. [$y = 5 - 2x = -2x + 5$ is of the form $y = ax + b$ with slope $a = -2$ and y-intercept $b = 5$]
3. Answer: (A) $\sqrt{13}$. [13 is not a perfect square; $\sqrt{36} = 6$, $2.75 = 11/4$ and $0.454545... = 5/11$ are all rational]
4. Answer: (D) 23. [$(x + 1/x)^2 = x^2 + 2 + 1/x^2$, so $x^2 + 1/x^2 = 25 - 2 = 23$]
5. Answer: (B) 14 cm. [The longest chord is a diameter = $2 \times 7 = 14$ cm]
6. Answer: (C) $1/2$. [Primes on a die : 2, 3, 5, so probability = $3/6 = 1/2$]

Section B – Very Short Answer Questions (2 marks each)

[10]

7. Quadrant / axis
 - (i) $(-5, 3)$: x-coordinate negative, y-coordinate positive, so it lies in Quadrant II. (1 mark)
 - (ii) $(0, -4)$: x-coordinate is 0, so it lies on the y-axis (on its negative side, 4 units below the origin). (1 mark)
8. Taxi fare
Fare for x km : $p(x) = 12x + 40$ (1 mark)
For 7 km : $p(7) = 12 \times 7 + 40 = 84 + 40 = ₹ 124$ (1 mark)
9. Rational and irrational number between 2 and 3
Rational number : $5/2 (= 2.5)$ (1 mark)
Irrational number : $\sqrt{5}$, since $4 < 5 < 9$ gives $2 < \sqrt{5} < 3$, and 5 is not a perfect square. (1 mark)
(Any other valid rational and irrational numbers, e.g. 2.4 and $\sqrt{6}$, may be accepted.)
10. 103×97
 $103 \times 97 = (100 + 3)(100 - 3)$ (1 mark)
 $= 100^2 - 3^2$ [(a + b)(a - b) = $a^2 - b^2$]
 $= 10000 - 9 = 9991$ (1 mark)
11. Centre of the circle through A, B, C
Join AB and BC. Draw the perpendicular bisector of AB and the perpendicular bisector of BC. (1 mark)
The point O where these two bisectors meet is the circumcentre, equidistant from A, B and C. The circle with centre O and radius OA passes through A, B and C. (1 mark)

Section C – Short Answer Questions (4 marks each)

[8]

12. Expand : $(3p + 2q - 4r)^2$
Using $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$, with $a = 3p$, $b = 2q$, $c = -4r$ (1 mark)
 $= (3p)^2 + (2q)^2 + (-4r)^2 + 2(3p)(2q) + 2(2q)(-4r) + 2(-4r)(3p)$ (2 marks)
 $= 9p^2 + 4q^2 + 16r^2 + 12pq - 16qr - 24pr$ (1 mark)

OR

$$\begin{aligned}x + y &= 9 \text{ and } xy = 20 \\x^2 + y^2 &= (x + y)^2 - 2xy \text{ (1 mark)} \\&= 9^2 - 2(20) = 81 - 40 = 41 \text{ (1 mark)}\end{aligned}$$

$$(x - y)^2 = (x + y)^2 - 4xy \quad (1 \text{ mark})$$

$$= 81 - 80 = 1 \quad (1 \text{ mark})$$

13. Chord of length 16 cm in a circle of radius 10 cm

Let O be the centre and AB the chord, AB = 16 cm. Draw OM $\perp$ AB. (1 mark)

The perpendicular from the centre to a chord bisects the chord, so AM = 16/2 = 8 cm. (1 mark)

In right $\triangle OMA$: $OM^2 = OA^2 - AM^2 = 10^2 - 8^2 = 100 - 64 = 36$ (1 mark)

OM = 6 cm. The chord is 6 cm from the centre. (1 mark)

OR

Chords AB and CD of a circle of radius 13 cm

AB is 5 cm from the centre O; the perpendicular OM bisects AB. (1 mark)

$AM^2 = 13^2 - 5^2 = 169 - 25 = 144$, so AM = 12 cm and AB = $2 \times 12 = 24$ cm. (1 mark)

For CD, at distance 12 cm : half-chord² = $13^2 - 12^2 = 169 - 144 = 25$ (1 mark)

Half-chord = 5 cm, so CD = 10 cm. (1 mark)

Section D – Case-Based Question (6 marks)

[6]

14. Cards numbered 1 to 20

(i) There are 20 equally likely outcomes (the numbers 1, 2, 3, ..., 20). (1 mark)

(ii) Multiples of 5 : 5, 10, 15, 20, which are 4 favourable outcomes. (1 mark)

$$P(\text{multiple of 5}) = 4/20 = 1/5 \quad (1 \text{ mark})$$

(iii)(a) Prime numbers from 1 to 20 : 2, 3, 5, 7, 11, 13, 17, 19, which are 8 numbers.

$$P(\text{prime}) = 8/20 = 2/5 \quad (1 \text{ mark})$$

(iii)(b) Numbers that are prime or a multiple of 5 : 2, 3, 5, 7, 10, 11, 13, 15, 17, 19, 20, which are 11 numbers (5 is counted once). (1 mark)

Numbers that are neither = $20 - 11 = 9$ (namely 1, 4, 6, 8, 9, 12, 14, 16, 18).

$$P(\text{neither prime nor a multiple of 5}) = 9/20 \quad (1 \text{ mark})$$